

ESQUIMALT HARBOUR, BC, Canada

WMO: 717980

Lat: 48.432N

Lon: 123.439W

Elev: 10

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	28.0	31.1	14.6	11.5	32.9	20.5	15.4	34.8	27.6	47.2	23.8	44.8	9.8	20	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	71.8	60.3	68.9	59.3	66.4	58.4	62.0	68.8	60.7	66.6	59.5	64.7	8.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.1	75.0	64.1	57.9	71.8	62.7	56.9	69.1	61.6	27.7	69.1	26.8	66.8	26.0	65.0	69.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.9	18.9	16.6	DB	24.8	80.0	3.7	3.2	22.2	82.3	20.0	84.2	17.9	86.0	15.3	88.4
			WB	22.7	65.3	4.4	1.7	19.5	66.5	17.0	67.5	14.5	68.5	11.4	69.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	41.9	42.4	44.8	48.5	53.5	57.3	60.6	60.4	56.7	50.5	45.2	41.5
	DBStd	8.05	4.94	4.33	4.18	3.97	4.24	3.70	3.25	3.12	3.30	3.52	4.62	4.78
	HDD50	1172	252	214	167	72	15	0	0	0	0	36	152	264
	HDD65	5370	716	634	627	495	358	233	140	145	250	449	594	729
	CDD50	1291	1	0	5	27	122	220	330	324	201	52	8	1
	CDD65	12	0	0	0	0	0	3	4	4	1	0	0	0
	CDH74	39	0	0	0	0	4	13	10	10	2	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
	WSAvg	6.7	7.1	7.2	7.2	7.1	7.2	7.8	6.9	6.0	5.1	5.6	6.5	6.9
	PrecAvg	25.4	4.2	2.9	1.9	1.3	1.0	0.7	0.6	0.8	1.3	2.4	3.9	4.4
(15) Precipitation	PrecMax	34.4	8.5	6.7	4.3	2.8	2.0	2.7	1.5	2.6	2.8	8.1	7.5	10.4
	PrecMin	15.3	0.6	0.4	0.5	0.7	0.2	0.1	0.0	0.0	0.2	0.2	0.6	0.4
	PrecStd	4.9	2.1	1.5	1.1	0.6	0.5	0.6	0.4	0.7	0.7	1.8	1.7	2.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	54.0	58.8	65.3	71.7	75.1	75.2	75.0	71.8	63.3	57.0	53.1
		MCWB	49.9	47.5	50.0	52.5	57.2	60.4	61.7	62.2	59.9	53.8	53.4	50.5
	2%	DB	51.1	51.3	54.5	59.4	65.3	68.8	71.3	70.6	66.9	59.2	54.3	50.6
		MCWB	48.4	47.0	48.5	50.9	54.8	58.3	60.6	60.7	58.5	53.6	51.5	48.1
	5%	DB	49.5	49.7	52.2	56.3	61.6	65.0	68.3	67.7	63.9	57.2	52.4	49.0
		MCWB	47.3	46.1	47.6	49.9	54.0	56.9	59.6	59.9	57.4	53.1	50.2	46.7
	10%	DB	48.1	48.2	50.4	54.0	59.0	62.3	65.9	65.3	61.5	55.5	51.1	47.6
		MCWB	46.2	45.2	46.8	48.8	52.9	55.5	58.7	58.9	56.6	52.3	49.1	45.5
	0.4%	WB	51.3	50.3	52.0	54.5	59.0	62.2	64.1	63.8	62.0	57.7	54.5	51.4
		MCDB	52.4	52.0	54.9	60.7	67.1	71.8	71.5	71.1	68.3	59.5	55.9	52.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.3	48.5	50.1	52.6	56.5	59.4	61.9	61.9	60.0	55.4	52.3	49.0
		MCDB	50.4	50.3	52.7	57.4	62.9	66.2	68.7	68.3	65.0	57.2	53.4	50.2
	5%	WB	47.9	47.1	48.7	51.1	54.9	57.7	60.5	60.6	58.4	53.9	51.0	47.3
		MCDB	49.1	48.9	51.4	54.9	60.1	63.4	66.4	66.3	62.2	55.7	52.1	48.5
	10%	WB	46.5	45.9	47.4	49.8	53.6	56.4	59.3	59.4	57.2	52.7	49.6	46.0
		MCDB	47.9	47.7	50.1	53.2	58.1	61.3	64.7	64.2	60.4	54.7	50.8	47.3
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.4	9.3	10.2	10.8	10.8	12.3	11.8	10.8	9.0	7.8	7.0
		MCDBR	8.4	10.3	11.4	14.2	15.4	15.7	16.2	15.4	14.6	11.0	8.4	7.8
		MCWBR	6.3	6.8	6.8	7.5	7.2	7.0	7.4	7.2	7.7	6.8	6.2	6.2
	5% WB	MCDBR	7.8	8.1	10.4	12.3	13.7	14.4	15.0	14.3	13.0	9.3	7.4	7.3
		MCWBR	6.4	6.3	6.8	7.1	7.0	7.2	7.8	7.4	7.7	6.7	6.0	6.3
(40) Clear-Sky Solar Irradiance	taub		0.320	0.319	0.334	0.362	0.382	0.360	0.351	0.356	0.342	0.335	0.328	0.316
	taud		2.516	2.543	2.514	2.425	2.388	2.468	2.530	2.515	2.553	2.557	2.496	2.484
	Ebn at Noon		239	269	281	282	279	286	287	281	275	259	233	224
	Edh at Noon		20	23	28	34	36	34	31	30	27	23	20	18
(44) All-Sky Solar Radiation	RadAvg		295	576	899	1360	1738	1901	2077	1752	1249	687	353	237
	RadStd		39	77	105	127	137	180	145	113	102	75	54	31

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	-0.99	-1.24	N/A	N/A	N/A	N/A
(47) Regional (76 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page